

Electrician

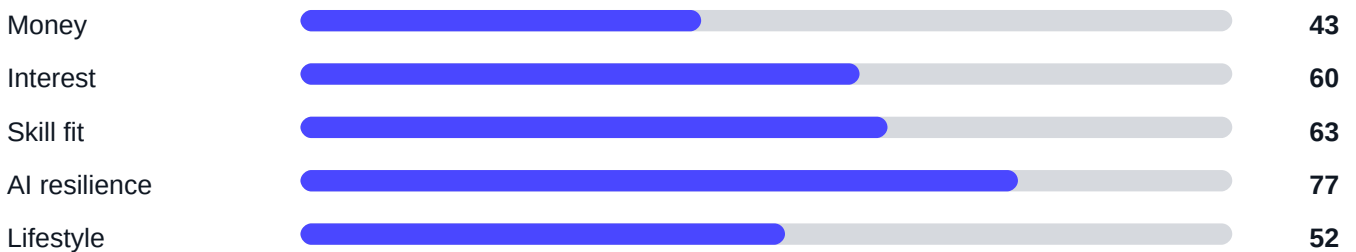
"You spend your day bending conduit, pulling wire, and reading prints in unfinished buildings, getting paid to make sure nothing you touched ever burns a house down or kills someone."

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Promising — go in with eyes open

Electrician could work: your strongest signal is AI resilience (77/100), your biggest watch-out is pay (43/100). You missed the AI coworker's mistake this round; that judgment is the skill to build before you rely on AI here.

Reality Score breakdown



The reality snapshot

Median pay: \$62,350/yr (2024) · Entry level: \$39,430 · Job outlook: +9% (2024-2034)

AI exposure: 18/100. Very low direct automation risk. The core work is physical, on-site, and requires a licensed human to legally sign off on safety. AI and robotics speed up design, estimating, and prefab, but someone still has to crawl the attic and make the connection. If anything, electrification (EVs, solar, data centers, grid upgrades) is expanding demand faster than AI is shrinking it.

The part nobody tells you: The romantic part (troubleshooting, finishing a clean panel) is maybe 20% of it. The other 80% is repetitive, body-grinding labor: drilling hundreds of identical holes, pulling stiff wire through long conduit runs, working overhead until your shoulders burn, kneeling on concrete, and being filthy, hot, or freezing. Add the unglamorous grind of memorizing and applying the National Electrical Code, and the years of low apprentice pay before the money shows up.

Your money plan

Chosen path: Public university (in-state) (4 years, \$46,660 tuition) · Region: US national average
Estimated debt after aid + part-time work: \$662 · Starting salary: \$39,430 · Break-even: 4.1 years
Low-risk money path — debt stays small relative to what you'll earn early on.

Your next 30 days

1. Lock down the fundamentals

Spend the first week mastering Ohm's Law, series/parallel circuits, and wire-gauge-to-ampacity (14/15A, 12/20A, 10/30A). Use free Khan Academy circuit lessons or Mike Holt's free basic videos.

2. Buy and use a multimeter

Get a cheap (~\$30) multimeter and a non-contact voltage tester. Practice measuring voltage, continuity, and resistance on batteries, outlets, and a lamp cord (unplugged) until it's second nature.

3. Build a mini wiring project

With a \$25 wiring kit or hardware-store parts and a length of NM cable, wire a single-pole switch to a light, then a GFCI receptacle, then a 3-way switch pair on a board. This is your hands-on proof you actually like the work.

4. Read the apprenticeship landscape

Compare a union JATC (NECA-IBEW Electrical Training Alliance, tuition-free + paid) vs. a merit-shop IEC apprenticeship in your state. Note application windows, aptitude test (math + reading), and physical requirements.

5. Knock out the math gate

Most apprenticeship aptitude tests are algebra and reading comprehension. Spend a week on the free Electrical Training Alliance / IBEW aptitude prep and basic algebra (fractions, ratios, solving for a variable).

6. Make real contact

Email or walk into 3 local electrical contractors and 1 JATC/IEC office. Ask about apprenticeship openings, pre-apprenticeship programs, and whether you can shadow a crew for a day. Getting on a real jobsite once beats a month of videos.

Verify it yourself

Primary source — U.S. Bureau of Labor Statistics:

<https://www.bls.gov/ooh/construction-and-extraction/electricians.htm>